

Making Magic with 3D Volume Style Transfer

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Figure 1: The effects of Magnifico's magic cauldron achieved by leveraging Volumetric Neural Style Transfer. © 2024 Disney

ABSTRACT

In order to achieve the unique look of a hand-drawn, 2D visual style blended with 3D effects elements in Walt Disney Animation Studios' film "Wish", effects artists leveraged new applications of Volumetric Neural Style Transfer (VNST) techniques. VNST, a method for stylizing 3D volume simulations by extracting the textures and shapes from a 2D image, has been increasingly more effective in assisting the stylization of effects through a number of the studio's previous projects. For "Wish", the effects artists took advantage of new advancements to inform and accelerate the visual development process, increase the reuse and efficiency of VNST throughout production, and achieve levels of style with detail unachievable with previous methods.

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1 INTRODUCTION

The popularity of style transfer techniques for 3D animation applications continues to grow as advancements in the field have made the techniques more robust and easier to leverage. Transport-Based Neural Style Transfer (TNST) [Kim et al. 2019] is a method for

applying textures and shapes from a 2D image onto a 3D volume. We utilized this approach in collaboration with Disney Research Studios (DRS) as a means to increase the expressiveness of previous Disney features such as "Raya and The Last Dragon" [Navarro and Rice 2021] and "Strange World" [Navarro 2023].

While VNST techniques garnered great success on these previous films, we needed a number of improvements in terms of workflow and implementation in order to scale to the needs of "Wish". Because VNST easily becomes computationally expensive, artistic workflows are often hindered due to complicated preprocessing of data. There were also stages of production where we still had not started to utilize VNST nor had we seen how far we could push its capabilities.

For "Wish", there were three key ways we wanted to expand upon our previous projects. First, because we knew we were targeting a unique look, we wanted to introduce the use of VNST earlier in our visual development process in order to inform and accelerate the initial stages. In order to increase our overall efficiency with VNST, we targeted the use of new innovations coming out of DRS, specifically with the use of a new *omniview* camera technique that had the potential to greatly increase our efficiency through asset reuse [Aurand et al. 2022]. Finally, we wanted to push the boundaries of what we have done with VNST by producing effects with levels of detail unobtainable by any previous method.

2 MAGNIFICO'S LAIR

While the new VNST workflows were applied throughout the film, the combined application of techniques in Magnifico's lair make this set a perfect showcase.

2.1 Early Exploration

When starting visual development, the goal in designing the many magical elements throughout the film was to blend the charm

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of traditional 2D animation with the precise look of a computer-generated simulation. Using a Houdini Pyro simulation as our volumetric base alongside its velocity field, we introduced a 2D style image that captured the traditional animation look. All of these were fed as inputs to the VNST system. Figure 2 showcases the output of one of these early tests. Inspired by these initial explorations, our 2D effects designers integrated these innovative shapes into further design work, which in turn informed subsequent stylized simulations.

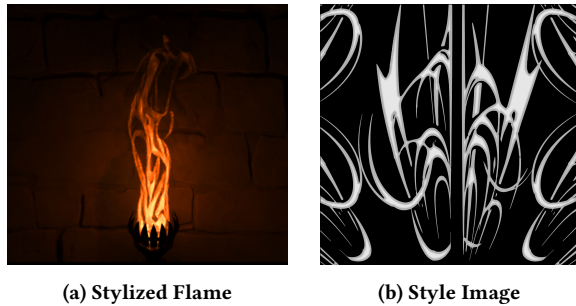


Figure 2: Flame stylizing test. © 2024 Disney

2.2 Omniview Camera

Leveraging the innovations from Disney Research [Aurand et al. 2022], we introduced omniview, a multi-camera optimization framework that addresses the limitations of single camera 2D style transfer methods. Historically, style transfer for volumetric data has relied on optimizing for a single view, utilizing a 2D to 3D pipeline that facilitates the re-sculpting of input volumes with textures and shapes extracted from an input image. This process involves rendering the 3D volume into a 2D data array for neural optimization, treating it as a content image, and then re-expanding the stylized output to 3D from a specific camera perspective, which limits viewing flexibility and the aesthetic range.

Despite the effectiveness of single-camera transfers for less demanding scenarios, the omniview approach significantly enhances both visual and computational outcomes. It improves visual quality across perspectives by leveraging the multidimensionality of volumetric content, resulting in more realistic and visually compelling textures, shapes, and patterns that appear naturally rotated and skewed, mirroring real-world appearances. Figure 3 showcases the effects we are able to achieve by utilizing eight omniview cameras when compared to only one. This was utilized throughout shots set in Magnifico’s lair to style the many magical smoke effects in a way that matched the overall look and feel of the film.

In addition to visual richness, this approach enhances pipeline efficiency by reducing preprocessing for VNST, notably through the omniview camera architecture, which allows for multi-perspective stylization. This improvement facilitates broader applications with less effort and enables artists of all experience levels to utilize these outputs without mastering the entire VNST pipeline. In summary, omniview has significantly advanced volumetric style transfer for visual effects by offering superior results, streamlining efficiency through stylized asset reuse, optimizing artist time, and reducing computational and storage demands.

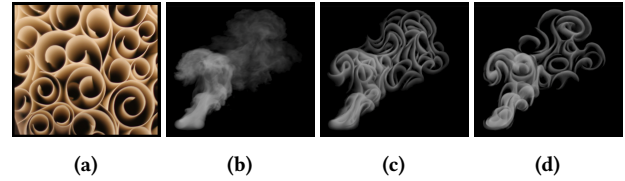


Figure 3: Smoke simulation stylized with omniview. (a) Style Image. (b) Input source. (c) Single camera VNST. (d) Multi-camera VNST utilizing eight omniview cameras. © 2024 Disney

2.3 Uniting 2D and 3D

At a broader scale, the effects assets built for visual development and the ones leveraging the omniview camera are relatively simple. In visually impactful shots, our aim was to push the limits of what we have been able to achieve with VNST. One such shot was the magic mixing of Magnifico’s concoction brewed in his cauldron that served as a highlight of his villainous turn.

For this scene, our process began with 2D effects animation depicting magic flowing into the cauldron. This animation was extruded and transitioned into 3D to accurately represent it within the appropriate spatial dimensions. These extruded shapes were then used as guides to generate a velocity field that helped drive a dry ice smoke simulation in Houdini. The volume output of this simulation was then stylized with VNST using its velocities and the Magnifico pattern. The final shot as seen in Figure 1, showcases the blended results of VNST, the 2D elements, and the simulation with VNST ensuring stylistic unity between 2D and 3D components.

3 CONCLUSION AND FUTURE WORK

As the efficiency and versatility of VNST techniques continue to improve with innovations such as the omniview camera technique and as our artists grow more skilled and comfortable leveraging such techniques in our films, the quality of our effects work grows stronger. In the future, we hope to explore integrating multiple styles within a single VNST process and optimize our use of omniview to broaden its application. Throughout the production process with VNST, artists were given the opportunity to leverage a powerful new tool which empowered them to elevate their artistic expression and breath unique life into our films now and going forward.

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